

IML 2026

Machine Learning Laboratory

LAB 04

Linear Regression from Scratch

Instructions

1. Complete the laboratory in Google Colab.
2. Use the dataset provided below.
3. NumPy, Pandas, and Matplotlib may be used.
4. `train_test_split()` may be used for splitting the data.
5. `StandardScaler()` may be used for feature scaling.
6. Linear Regression must be implemented from scratch.
7. Do not use a ready-made Linear Regression model for the main implementation.
8. Implement training using Gradient Descent.
9. Include all required outputs, calculations, and plots.
10. Compare your implementation with a standard library implementation only for verification.
11. Save the notebook using the required filename and upload it to Google Classroom.

1. Dataset

Use the **California Housing Dataset**.

The dataset is available through the following source:

https://scikit-learn.org/stable/modules/generated/sklearn.datasets.fetch_california_housing.html

Use the available numerical features to predict the target variable.

2. Task 1 – Load and Prepare the Data

1. Load the dataset and create the feature matrix X and target vector y .
2. Display:
 - Number of samples
 - Number of features
 - Feature names
 - First five observations
3. Split the dataset into training and testing sets using:

```
test_size = 0.2,    random_state = 42
```

4. Standardize the input features using `StandardScaler()`.

3. Task 2 – Linear Regression Model

Implement Multiple Linear Regression from scratch.

The prediction function should follow:

$$\hat{y} = Xw + b$$

where:

- X is the input feature matrix,
- w is the weight vector,
- b is the bias,
- $\hat{y}$ is the predicted output.

Initialize the weights and bias and implement the prediction calculation using NumPy.

4. Task 3 – Mean Squared Error

Implement the Mean Squared Error (MSE) cost function:

$$J(w, b) = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

where N is the number of training samples.

Calculate and display the cost during training.

5. Task 4 – Gradient Descent

Train the Linear Regression model using Gradient Descent.

Use the gradients:

$$\frac{\partial J}{\partial w} = -\frac{2}{N} X^T (y - \hat{y})$$

$$\frac{\partial J}{\partial b} = -\frac{2}{N} \sum_{i=1}^N (y_i - \hat{y}_i)$$

Update the parameters using:

$$w \leftarrow w - \eta \frac{\partial J}{\partial w}$$

$$b \leftarrow b - \eta \frac{\partial J}{\partial b}$$

where η is the learning rate.

1. Train the model for a suitable number of epochs.
2. Experiment with the learning rate.
3. Store the MSE after each epoch.
4. Plot the training MSE against the number of epochs.

6. Task 5 – Prediction and Evaluation

Use the trained parameters to predict the target values for the test set.

Calculate:

1. Mean Squared Error (MSE)
2. Root Mean Squared Error (RMSE)
3. R^2 score

The coefficient of determination is:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

Report the obtained evaluation values.

7. Task 6 – Visualization

Create the following plots:

1. Training MSE vs. Epochs.
2. Actual values vs. Predicted values for the test set.

Briefly comment on the observed relationship between the actual and predicted values.

8. Task 7 – Verification

For verification only, train a Linear Regression model using a standard machine learning library.

Compare:

- MSE
- RMSE
- R^2

between your implementation and the library implementation.

Note: The library model must not be used as a replacement for the from-scratch implementation.

9. Implementation Restrictions

Allowed:

- NumPy
- Pandas
- Matplotlib
- `train_test_split()`
- `StandardScaler()`
- Standard library functions for evaluation

Must be implemented from scratch:

- Prediction function

- MSE cost function
- Gradient calculation
- Weight update
- Bias update
- Gradient Descent training loop

Do not use for the main implementation:

- `LinearRegression()`
- Any other ready-made regression algorithm

10. Submission

Complete the assignment in Google Colab.

Ensure that:

- All code cells are executed.
- Required outputs and plots are visible.
- The notebook contains the complete from-scratch implementation.

Required File Name:

`IML2026_Roll-No_Lab_04.ipynb`